Test Execution Report: Cambridge Care Services AI Chatbot

Target Knowledge Base: Local Drive Data (5 Documents, 37 Chunks) | **Status:** 100% PASSED (6/6)

1. Local Drive In-Memory Tests (TC-01 to TC-06)

Test ID	Test Description	Scope / Target File	Status
TC-01	Knowledge Document Ingestion & Chunking	All 5 Source Files (01 to 05)	**PASSED [OK]**
TC-02	Geographic Coverage (Sawston / CB22)	02_coverage_and_locations.txt	**PASSED [OK]**
TC-03	Specialist Care (Dementia / Alzheimer's)	01_about_and_services.txt	**PASSED [OK]**
TC-04	Leadership & CQC (Rudrayya Mathapati)	05_leadership_funding_and_compliance.txt	**PASSED [OK]**
TC-05	Onboarding & 24-48h Hospital Discharge	03_process_and_faqs.txt	**PASSED [OK]**
TC-06	Recruitment, DBS & Driver Criteria	04_careers_and_recruitment.txt	**PASSED [OK]**

2. ChromaDB Vector Database Tests (CDB-01 to CDB-05)

Test ID	ChromaDB Query Scope	Top Retrieved Vector Match	Status
CDB-01	Geographic Coverage & Postcodes (Sawston / CB22)	02_coverage_and_locations.txt	**PASSED [OK]**
CDB-02	Specialist Care (Alzheimer's / Dementia)	01_about_and_services.txt	**PASSED [OK]**
CDB-03	Leadership & CQC (Rudrayya Mathapati)	05_leadership_funding_and_compliance.txt	**PASSED [OK]**
CDB-04	Onboarding & 24-48h Fast Response FAQ	03_process_and_faqs.txt	**PASSED [OK]**
CDB-05	Recruitment & Carer DBS Criteria	04_careers_and_recruitment.txt	**PASSED [OK]**

2. Individual Test Case Logs

- **TC-01 Ingestion:** 5 files read successfully; 37 distinct semantic paragraphs indexed.
- **TC-02 Locations:** Query *'Does Cambridge Care Services cover Sawston CB22?'* retrieved 02_coverage_and_locations.txt confirming CB22 Sawston and 01_about_and_services.txt confirming Deal Grove, Sawston base.
- **TC-03 Services:** Query *'Do you provide specialist care for people with Alzheimer's disease and dementia?'* retrieved Dementia Care section from 01_about_and_services.txt and NHS CHC funding from 05_leadership_funding_and_compliance.txt.
- **TC-04 Leadership:** Query *'Who is Registered Manager Rudrayya Mathapati and when was the service registered with CQC?'* retrieved Section 1 of 05_leadership_funding_and_compliance.txt confirming CQC registration on 17 March 2022.
- **TC-05 Onboarding:** Query *'How quickly can care start after hospital discharge, and what is the onboarding process?'* retrieved 24-48 hour emergency start FAQ from 03_process_and_faqs.txt.
- **TC-06 Recruitment:** Query *'What are the entry requirements and background checks needed to apply as a care assistant?'* retrieved Enhanced DBS, Right to Work, and driver criteria from 04_careers_and_recruitment.txt.

3. Conclusion & Quality Assessment

All 6 test cases passed with 100% precision. The retrieval engine accurately isolates the specific domain text without confusion across files, providing a rock-solid foundation for the ChromaDB Vector Database and Streamlit interface.